EMPOWERMENT TECHNOLOGIES

ACADEMIC

Grade 11/12

Curriculum Guide

EMPOWERMENT TECHNOLOGIES
Pre-requisite: None

Course Description:
This course equips learners with the knowledge and skills to maintain and optimize computer systems, secure data, and use digital productivity tools for effective communication, collaboration, and problem-solving. It also introduces emerging technologies such as Artificial Intelligence (AI), Blockchain, and Internet of Things (IoT), enabling learners to develop practical solutions, strengthen ethical awareness, and gain experience in no-code AI development and web development for creating AI models, applications, and functional websites.

Elective: Academic
Time Allotment: 80 hours for one term

CONTENT	**CONTENT STANDARDS** *The learners demonstrate an understanding of…*	**LEARNING COMPETENCIES** *The learners…*
GREEN ICT AND SUSTAINABLE COMPUTING PRACTICES **Introduction to Green ICT** • E-waste management • Energy-efficient devices **Sustainable Computing Practices** • Cloud-based solutions • Eco-friendly gadgets/computing devices **Role of ICT in Climate Change Mitigation**	maintaining sustainable, efficient, and secure computer systems, emphasizing Green ICT principles, effective hardware and software maintenance, system optimization, data management, and cybersecurity measures.	1. discuss the principles and practices of Green Information and Communications Technology (ICT) and sustainable computing.

<table>
<tr><th>CONTENT</th><th>CONTENT STANDARDS

The learners demonstrate an understanding of…</th><th>LEARNING COMPETENCIES

The learners…</th></tr>
<tr><td>MAINTENANCE AND OPTIMIZATION OF COMPUTER SYSTEMS

Hardware Maintenance
Importance of hardware maintenance
External cleaning of computer components and peripherals
Software Maintenance
Updating operating systems and application software
Managing applications
installations and uninstallations
startup applications
System Optimization
Utility tools
defragmentation
disk cleanup
check disk</td><td rowspan="2"></td><td>2. perform basic maintenance and optimization of computer systems.</td></tr>
<tr><td>Digital File Management
File extensions and types (e.g., .docx, .pdf, .xlsx)
File organization (e.g., naming conventions)
Cloud storage services (e.g., Google Drive, OneDrive)</td><td>3. perform management of digital files.</td></tr>
</table>

CONTENT	CONTENT STANDARDS *The learners demonstrate an understanding of...*	LEARNING COMPETENCIES *The learners...*
Data Backup and Recovery • Backing up using local and cloud storage solutions • Restoring data from backups		4. perform procedures in data backup and recovery.
Security Measures • Antivirus software installation and maintenance • Firewall settings and other security protocols • Two-Factor Authentication (2FA) and Multi-Factor Authentication (MFA) • Email Masking • Incident response and reporting		5. apply security measures to protect computer systems and data from cyber threats.
Performance Standard: The learners apply sustainable and efficient ICT practices in maintaining, optimizing, and securing computer systems to minimize environmental impact, ensure data integrity, and enhance system performance through proactive hardware and software management.		
ADVANCED WORD PROCESSING SKILLS **Advanced Formatting and Layouts** • Style and themes application • Section breaks for complex documents layouts • Table of contents, indexes, and references • Formatting page number (Roman numerals,	advanced features and functionalities of digital productivity tools, including word processing, spreadsheets, and presentation software, in creating complex, dynamic, and accessible content for effective	6. apply advanced word processing skills in creating, formatting, and managing complex documents.

CONTENT	CONTENT STANDARDS *The learners demonstrate an understanding of...*	LEARNING COMPETENCIES *The learners...*
different section numbering) **Multimedia Integration** • Multimedia and other interactive elements • Linking to external data sources or integrating live data **Accessibility Features** • Accessibility checkers • Alternative text to images **Collaboration Features** • Using track changes and comments • Real-time co-authoring **Security and Document Management** • Applying password protection • Using digital signatures and encryption	communication, collaboration, and problem-solving.	
ADVANCED SPREADSHEET SKILLS ADVANCED SPREADSHEET FUNCTIONS **Logical functions** • IF • AND • OR **Lookup functions** • VLOOKUP		7. apply advanced spreadsheet skills in analyzing, manipulating, and visualizing complex data.

CONTENT	CONTENT STANDARDS *The learners demonstrate an understanding of...*	LEARNING COMPETENCIES *The learners...*
• LOOKUP • INDEX/MATCH **Text manipulation functions** • CONCATENATE • TEXT • LEFT • RIGHT • MID **Date and time functions** • TODAY • NOW • DATE • TIME **Data Analysis Tools** • Creating pivot tables • Applying conditional formatting with combination of rules • Performing hat-If Analysis (Goal Seek, Scenario Manager) • Using solver for optimization problems (add in) **Data Validation and Protection** • Applying data validation		

CONTENT	CONTENT STANDARDS *The learners demonstrate an understanding of…*	LEARNING COMPETENCIES *The learners…*
• Protecting cells, sheets, or workbooks **Visualization and Charts**: • Creating advanced charts (e.g., combo charts and waterfall charts) • Using sparklines • Developing data-driven visuals **Collaboration and Integration** • Sharing and collaborating on spreadsheets • Linking spreadsheets to external data sources (e.g., databases or web data)		
ADVANCED PRESENTATION SKILLS **Dynamic Visual Storytelling** ● Custom Animations and Motion Paths o Creating custom motion paths o Syncing animations with multimedia elements ● Multimedia Integration and Editing o Embedding and editing video and audio o Synchronizing multimedia elements with slide transitions or animations o Using 3D models and augmented reality elements **Efficient Design and Branding** ● Slide Master & Custom Templates		8. apply advanced presentation skills in creating visually engaging and interactive presentations.

CONTENT	CONTENT STANDARDS *The learners demonstrate an understanding of…*	LEARNING COMPETENCIES *The learners…*
○ Using slide Masters ○ Creating custom templates ○ Applying themes ● Data Integration & Advanced Charts ○ Integrating real-time data feeds ○ Creating advanced chart types ○ Developing pivot charts and graphs		
ADVANCED EMAIL MANAGEMENT SKILLS **Email Automation** • Using email marketing tools • Setting up auto responders and rules • Creating email templates **Email Security** • Encrypting emails • Using digital signatures • Applying Two-Factor Authentication (2FA) and Multi-Factor Authentication (MFA) **Email Collaboration and Integration** • Integrating email with productivity tools (e.g., calendars or task managers)		9. apply advanced email management skills to optimize productivity, security, and effectiveness in professional and personal contexts.

CONTENT	CONTENT STANDARDS *The learners demonstrate an understanding of...*	LEARNING COMPETENCIES *The learners...*
• Using email delegation **Email Organization** • Applying filters and rules • Archiving emails • Using keyboard shortcuts		
Performance Standard: The learners create documents and other digital outputs using advanced features and functionalities of digital productivity tools for effective communication, collaboration, and problem solving.		

ARTIFICIAL INTELLIGENCE
 	
Foundations of AI
• Introduction to AI
• History of AI
• Types of AI
• Key Components of AI
    ○ Data
    ○ Algorithms
    ○ Hardware
• AI Programming vs Traditional Programming
• AI Techniques and Algorithms
    ○ Machine Learning
    ○ Deep Learning
    ○ Natural Language Processing (NLP)
    ○ Computer Vision
	Artificial Intelligence (AI), Blockchain, IoT, and No-Code AI Development to foster data literacy, ethical awareness, and critical thinking.	10. discuss fundamental concepts of Artificial Intelligence (AI).

CONTENT	CONTENT STANDARDS *The learners demonstrate an understanding of…*	LEARNING COMPETENCIES *The learners…*

Applications of AI
• AI in Various Fields
  ○ Business
  ○ Education
  ○ Entertainment
  ○ Healthcare
  ○ Transportation
 	
• AI in Future Careers
  ○ AI-Driven career paths
  ○ AI in emerging technologies
  ○ Lifelong learning and AI certifications
  ○ Entrepreneurship in AI (AI startups and innovations)
		11. explain the applications of Artificial Intelligence (AI) in various fields and future careers.

AI Ethics and Social Implications • Importance of ethical AI practices • Responsible AI principles (fairness, transparency, accountability) • Beneficial and harmful uses of AI		12. discuss AI Ethics and social implications.

CONTENT	CONTENT STANDARDS *The learners demonstrate an understanding of...*	LEARNING COMPETENCIES *The learners...*

• Bias and fairness in AI systems
   o Algorithmic bias
   o Sources of bias in AI (data, human influence, algorithms)
   o Techniques for minimizing bias in AI models		

BLOCKCHAIN AND WEB3 TECHNOLOGIES **Blockchain Fundamentals** • Types of blockchains • Consensus mechanisms • Smart contracts • Decentralized applications (DApps) • Cryptography and security • Real-world applications **Web3 Technologies** • Decentralized Finance (DeFi) • Non-Fungible Tokens (NFTs) • Metaverse and Virtual Reality (VR) • Decentralized Autonomous Organizations (DAOs) • Interoperability and cross-chain Solutions		13. discuss concepts of Blockchain and Web3 Technologies.
Internet of Things (IoT) Integration • Introduction to IoT • Practical IoT Applications • IoT Security and Privacy Issues		14. discuss IoT systems, concepts, components, and applications.

<table>
<tr><th>CONTENT</th><th>CONTENT STANDARDS

The learners demonstrate an understanding of…</th><th>LEARNING COMPETENCIES

The learners…</th></tr>
<tr><td>NO-CODE AI DEVELOPMENT

Introduction to No-Code AI Development
• Definition of no-Code AI
• Differences between no-Code AI and traditional AI development
• Benefits and limitations of no-Code AI
• Overview of popular no-Code AI tools (e.g., Teachable Machine, Lobe, and Dialog flow)</td><td rowspan="3"></td><td>15. discuss the concepts of No-Code AI development.</td></tr>
<tr><td>No Code Machine Learning Models
• Choosing platform
• Exploring interfaces and features
• Data preparation and preprocessing
• Model building and training</td><td>16. develop machine learning models using no-code machine learning platforms.</td></tr>
<tr><td>No-Code Mobile App Development with AI Integration
• Integrating AI models into mobile apps
• Building AI-Powered apps (e.g., voice-to-text, chatbot app)
• Publishing and testing mobile apps</td><td>17. develop AI-integrated mobile applications using no-code platforms.</td></tr>
<tr><td colspan="3">Performance Standard:

The learners develop AI-driven models and applications while adhering to ethical guidelines and best practices.</td></tr>
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The learners...</th></tr>
<tr><td>NO-CODE WEB DEVELOPMENT
• Introduction to no-code web development and its benefits for entrepreneurs
• Identifying business needs and website goals
• Choosing the right no-code platform based on business requirements (e.g., Wix for simplicity and Webflow for flexibility)

Building a Business Website
• Planning website structure and user experience (UX)
• Designing an appealing and user-friendly interface (UI)
• Creating essential pages (e.g., homepage, about us, products/services, and contact page)
• Adding interactive elements (forms, maps, social media integration)
• Optimizing websites for search engines (SEO basics)
• Embedding AI Features
  o Adding chatbots or virtual assistants

E-commerce Integration
• Setting up an online store and product listings
• Integrating payment gateways securely
• Managing orders and customer service</td><td>No-Code web development to foster practical application skills for real-world innovation and future career opportunities.</td><td>18. develop functional websites using no-code web development tools to create an online platform for business ideas.</td></tr>
</table>

CONTENT	CONTENT STANDARDS *The learners demonstrate an understanding of...*	LEARNING COMPETENCIES *The learners...*
Showcasing a Business Idea • Creating compelling content (e.g., text, images, and videos) to showcase products/services • Building an online portfolio or gallery • Integrating testimonials and customer reviews **Publishing and Marketing** • Launching websites and registering domain name • Promoting websites through social media and online channels • Analyzing website traffic and user engagement		
Performance Standard: The learners develop functional websites using no-code development tools.		